AWS CLOUD LEARNING SERIES

Building an AWS VPC from Scratch:
Subnets, Route Tables, Internet Gateway and EC2 Deployment

A Step-by-Step Practical Guide to Creating a Custom IPv4 Network on Amazon Web Services

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Document 2

This guide takes you from a CIDR plan to a custom VPC, a routed subnet, an Internet Gateway, and an EC2 instance. The emphasis is on understanding how the pieces work together—not just following console clicks.

Practical learning guide

Console labels can change; verify current AWS interface wording as you work.

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1 Introduction

A VPC (virtual private cloud) is a logically isolated network that you configure in an AWS Region. This lab builds one small IPv4 environment: define a VPC address range, carve out a subnet, configure the routes that direct traffic, attach an Internet Gateway to the VPC, associate a route table with the subnet, and launch one EC2 instance into that subnet.

The objective is to reason about the network. A name such as "public subnet" is not a magical switch, and no single component guarantees internet reachability. Routes choose paths; IP addressing supplies endpoints; security controls and the host application determine what traffic is accepted.

By the end, you should be able to: explain a VPC and CIDR plan; create and verify a subnet in an Availability Zone; describe the local and default routes; attach an Internet Gateway and associate a route table; launch an instance into the intended VPC and subnet; and distinguish routing from traffic authorization.

2 Prerequisites and architecture

You should be comfortable with basic EC2 and IAM concepts, IPv4 addresses, and CIDR notation. You also need permission in an AWS account to create VPC and EC2 resources. The exact permissions depend on your organization's policies. AWS charges can apply to resources and public IPv4 addresses; check current pricing, instance availability, and any Free Tier eligibility before launching. Delete lab resources when finished if they are no longer needed.

This is a procedure, not a report of an executed deployment. No resource IDs, successful results, or screenshots are implied. Console options may vary by Region, account permissions, and AWS interface updates.

Table 1: One consistent example used throughout the lab.

Resource	**Example configuration**
VPC	`Learning-VPC`; `10.0.0.0/16`
Subnet	`Learning-Public-Subnet`; `10.0.1.0/24`; one chosen AZ
Route table	`Learning-Public-RT`; local route plus IPv4 default route
Internet Gateway	`Learning-IGW`; attached to `Learning-VPC`
EC2 instance	`Learning-EC2`; automatically assigned private address; request a public IPv4 address for this lab if appropriate

These are illustrative values, not universal defaults. Before using them, check that the ranges do not overlap with existing VPCs, connected networks, VPNs, or organizational address plans. Change names and ranges to fit your environment while preserving the same containment and non-overlap rules.

2.1 Architecture at a glance

The lab's intended relationships and an example of packet forwarding are shown separately below. The first diagram is a resource map, not a strict containment chain. The second describes a possible inbound web request and the conditions required for it to reach an application.

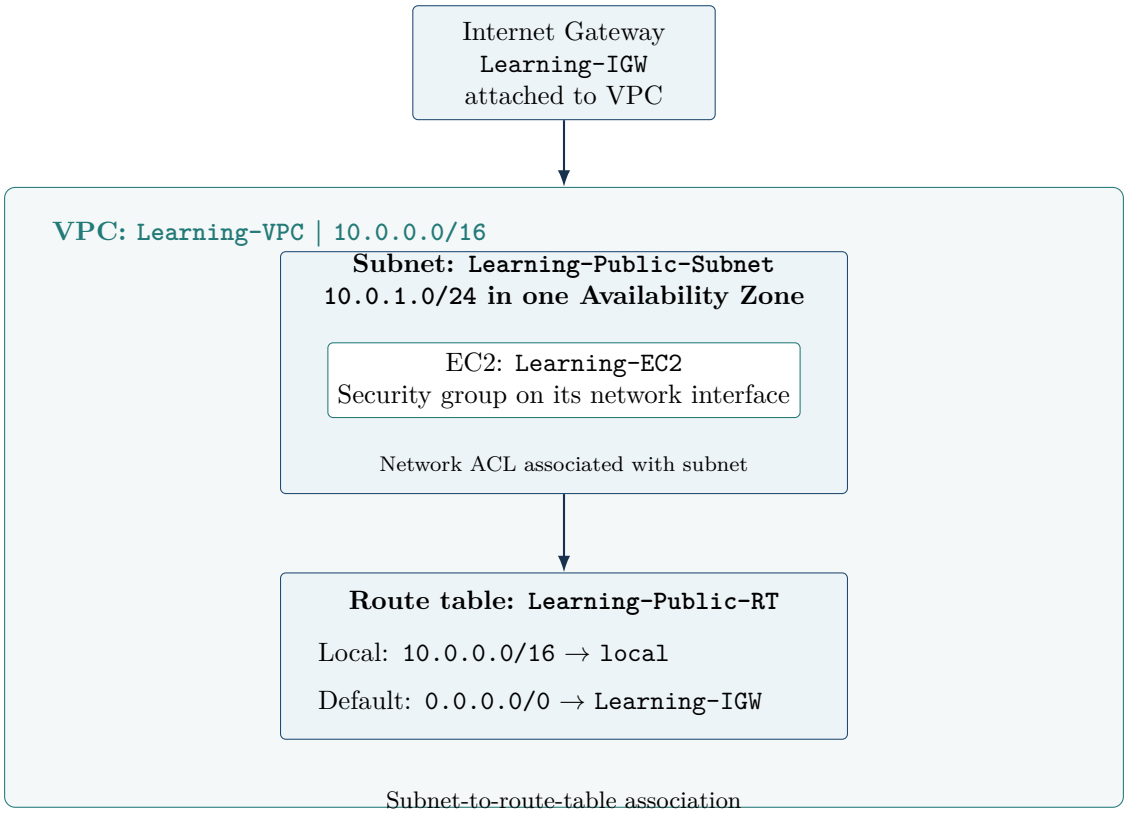


Figure 1: Resource relationships, not a containment chain. The subnet and route table are separate VPC resources; their association is labeled below the connector, and the Internet Gateway is shown attached to the VPC.

3 Choose the address ranges first

3.1 Reading a CIDR prefix

An IPv4 address has 32 bits. In a prefix such as `10.0.0.0/16`, the `/16` indicates that the first 16 bits identify the network portion; the remaining 16 bits are available to vary within that block. It therefore contains $2^{32-16} = 2^{16} = 65{,}536$ IPv4 addresses, from `10.0.0.0` through `10.0.255.255`.

For `10.0.1.0/24`, 24 bits identify the network portion and 8 bits remain. It contains $2^{32-24} = 2^8 = 256$ addresses, from `10.0.1.0` through `10.0.1.255`. A shorter prefix means a larger range: a /16 contains many /24-sized blocks.

Table 2: Nested example ranges inside the VPC.

Block	Range	Relationship
`10.0.0.0/16`	`10.0.0.0-10.0.255.255`	VPC range: 65,536 total addresses
`10.0.1.0/24`	`10.0.1.0-10.0.1.255`	Chosen subnet, entirely within VPC
`10.0.2.0/24`	`10.0.2.0-10.0.2.255`	Another possible, non-overlapping subnet
`10.0.3.0/24`	`10.0.3.0-10.0.3.255`	Another possible, non-overlapping subnet

A subnet CIDR must be inside the VPC's address space. Subnets in the same VPC cannot overlap, because an address would otherwise belong ambiguously to multiple subnet ranges. Overlap between

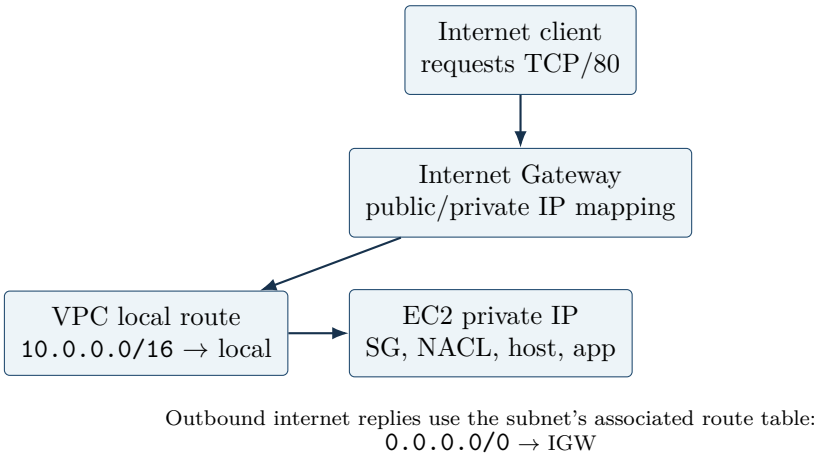


Figure 2: Conceptual inbound delivery uses local VPC routing after IGW address mapping. The default route to the IGW is relevant to internet-bound outbound traffic.

separately managed networks can also make routing ambiguous when they are later connected using peering, VPN, or other connectivity. Address planning now avoids painful redesign later.

Do not choose the largest range reflexively. A range must be large enough for expected resources and growth, but it must also fit an organization's larger plan and avoid conflicts with on-premises networks and other VPCs. Modern network planning uses CIDR ranges, not the obsolete classful A/B/C system as the primary sizing method.

3.2 AWS subnet addresses: total versus assignable

AWS reserves five IPv4 addresses in each ordinary IPv4 subnet: the first four addresses and the final address. For `10.0.1.0/24`, these are `10.0.1.0`, `10.0.1.1`, `10.0.1.2`, `10.0.1.3`, and `10.0.1.255`. Thus, the subnet has 256 total addresses and 251 available for assignment by AWS.

This is AWS's subnet rule; do not substitute the generic networking shortcut of subtracting a network address and broadcast address (which would yield 254). AWS reserves five, not two, in an ordinary subnet. AWS does not support broadcast within a VPC, but the last address is still reserved.

4 How the components fit together

A VPC defines a logically isolated virtual network and its configured IP ranges. A VPC spans the Availability Zones in a Region; an individual subnet belongs to exactly one AZ. You can create multiple non-overlapping subnets within the VPC.

A route table is a VPC resource containing destination-to-target rules. Each subnet is associated with exactly one route table at a time; one route table can serve multiple subnets. A VPC has a main route table. A subnet without an explicit association uses that main table. A custom table does not automatically govern a subnet just because both are in the same VPC.

An Internet Gateway attaches to the VPC. It is not attached directly to a subnet or instance. A route table associated with a subnet can contain a route whose target is that attached gateway. An EC2 instance is launched into a subnet and receives a network interface; security groups are associated with that interface/resource. Network ACLs (NACLs) apply at subnet level. These components work together

without being a simple parent-child chain.

4.1 The two routing entries

A route has a destination and a target. The VPC's local route is created automatically in route tables. For this example it is:

`10.0.0.0/16` ⟶ `local`

This route provides a path for traffic to the VPC's local IPv4 range. It does not override security groups, NACLs, or host controls; routed traffic can still be denied.

The route added for the lab is `0.0.0.0/0` ⟶ `Learning-IGW`. It is the default route for IPv4 destinations not matched by a more-specific route. Routers use longest-prefix matching: among matching routes, the most-specific prefix wins. This rule directs packets toward the IGW; it does not assign public addresses, authorize traffic, or guarantee that a service can be reached.

5 Lab: create the VPC

This guide uses the individual-resource workflow to make each networking component visible. AWS may also offer a "VPC and more" style workflow that creates supporting resources automatically; that is convenient but hides some of the relationships this exercise is designed to teach.

1. Sign in to the AWS Management Console and select the Region where you intend to work.
2. Open the VPC service and go to the VPC dashboard or VPC list.
3. Choose the create-VPC operation. Select the VPC-only/ VPC-only resource option if the wizard offers a choice between an individual VPC and a bundled setup. Labels can vary.
4. Enter name tag `Learning-VPC` and IPv4 CIDR `10.0.0.0/16`. The CIDR defines the IPv4 address space from which subnets will be carved.
5. Keep default tenancy for an ordinary learning lab unless you have a specific requirement. Tenancy is a high-level choice about the underlying host placement model; dedicated tenancy can affect cost and is not needed for this example.
6. Leave IPv6 unconfigured for this IPv4-focused exercise. IPv6 is a separate address family with its own CIDRs and routes; an IPv4 default route does not route IPv6.
7. Create the VPC. Then inspect the VPC list/details and verify the state is available and the primary IPv4 CIDR is `10.0.0.0/16`.

Creating a VPC alone does not create this lab's subnet or give it an Internet Gateway route. A nondefault VPC also should not be assumed to have the supporting internet resources of a default VPC.

6 Lab: create a subnet

A subnet is an address range inside a VPC in which you can place resources. Each subnet is associated with one Availability Zone (AZ), an isolated location within an AWS Region. Spreading subnets and

workloads across AZs can help design for AZ-level failures, but this single-subnet exercise does not build a multi-AZ architecture.

1. In the VPC console, open **Subnets** and choose the create-subnet operation.
2. Select `Learning-VPC`.
3. Set the subnet name to `Learning-Public-Subnet`.
4. Choose one AZ in the selected Region. Record the choice so you can verify the instance placement later.
5. Enter IPv4 CIDR `10.0.1.0/24`. Check that it is fully inside `10.0.0.0/16` and does not overlap any existing subnet in that VPC.
6. Create the subnet. Return to the subnet list/details and verify that it is available, has the intended CIDR, belongs to `Learning-VPC`, and is in the chosen AZ.

The `Public` part of the name is only a label. A subnet name does not configure routes or internet access. Creating a subnet does not attach an IGW, add a default route, or assign public addresses to instances. In this /24, AWS reserves five addresses, leaving 251 available for assignment by AWS.

7 Lab: create a custom route table

A route table acts as a traffic-direction list: for a packet's destination, it identifies the next target. The target may be local routing or a gateway. A route table affects routing for its associated subnet; it is not a firewall.

1. In the VPC console, open **Route tables** and choose create route table.
2. Name it `Learning-Public-RT`, and select `Learning-VPC` as its VPC.
3. Create the table, select it, and inspect its **Routes** tab.
4. Confirm that a local route exists for `10.0.0.0/16` with target `local`. The VPC's main route table is created automatically, but this new custom route table is not automatically selected for every subnet.

8 Lab: create and attach an Internet Gateway

An Internet Gateway is the VPC-level connection point used for internet traffic. Attaching one to a VPC does not, by itself, make every subnet or instance internet-reachable. Routing, addressing, and security controls still matter.

1. In the VPC console, open **Internet gateways** and choose create Internet Gateway.
2. Enter the name tag `Learning-IGW` and create it.
3. Select the new gateway and choose the attach-to-VPC operation.
4. Select `Learning-VPC` and attach it.
5. Verify the gateway's state shows it attached and inspect the associated VPC ID/name.

The IGW is not a general-purpose firewall and does not authorize all connections. It provides the gateway function; routes and traffic controls determine whether traffic can flow.

9 Lab: add the default IPv4 route

1. Open **Route tables**, select `Learning-Public-RT`, then open its **Routes** tab.
2. Choose the edit-routes operation and add a route.
3. Set destination to exactly `0.0.0.0/0`; select `Learning-IGW` as the target.
4. Save changes, then verify both routes are listed.

Table 3: Expected route entries in the custom route table.

Destination	Target	Meaning
`10.0.0.0/16`	`local`	Local route for the VPC IPv4 range
`0.0.0.0/0`	`Learning-IGW`	Default path for otherwise-unmatched IPv4 destinations

The default route directs eligible IPv4 packets toward the IGW. It does not create a public IP address, permit an inbound security-group rule, override a NACL or host firewall, or start an application. IPv6 would require its own IPv6 CIDR and routing configuration; it is outside this lab.

10 Lab: associate the route table with the subnet

This association is essential: the routes in the table determine paths for the associated subnet. The IGW is attached to the VPC, while the subnet uses a route table that contains a route targeting that IGW.

1. Select `Learning-Public-RT` in the route tables list.
2. Open **Subnet associations**, then choose the edit-associations operation.
3. Select `Learning-Public-Subnet` and save.
4. Verify the explicit association is shown for that subnet.

If no explicit custom-table association exists, the subnet uses the VPC's main route table. Do not infer which route table is active based on names; inspect the subnet association and the main-table indicator.

11 Lab: launch an EC2 instance

An EC2 instance is placed into a subnet when its network interface is created. Selecting the correct VPC and subnet ensures that the interface receives an address from the intended subnet and uses its routing association. Instance type and image availability, price, and eligibility vary; check the current console and pricing information before launch.

1. Open the EC2 console and choose **Launch instance**.
2. Name the instance `Learning-EC2`.

3. Choose a currently available, suitable Amazon Machine Image, such as an Amazon Linux image, and a small instance type appropriate for a learning lab.
4. If you plan to connect using SSH, create or select a key pair and store its private key securely. Do not put private keys or passwords in this document or share them.
5. Expand network settings. Select VPC `Learning-VPC` and subnet `Learning-Public-Subnet`.
6. For this lab, enable auto-assignment of a public IPv4 address if the launch wizard exposes the setting and direct IPv4 internet communication is appropriate. This nondefault subnet commonly has public-IP auto-assignment disabled by default; the launch option may override the subnet setting in supported configurations.
7. Create or select a security group. Allow only the inbound traffic needed for your test. For SSH, restrict TCP port 22 to your trusted public IP address where feasible; do not routinely open SSH to `0.0.0.0/0`. For a web test, allow TCP/80 or TCP/443 only as needed.
8. Review the configuration and launch the instance.
9. Wait for the instance to reach the appropriate running and status-check state. Inspect its VPC, subnet, private IPv4, public IPv4 (if requested), and security group in the instance details/networking view.

The private address is automatically selected from the subnet unless you deliberately specify a valid private address. A public IPv4 address and an associated subnet route to the IGW are both needed for ordinary direct IPv4 communication with the internet through the IGW. Actual inbound reachability additionally depends on security-group rules, NACL rules, host firewall settings, and a service listening on the requested port. A running instance is not automatically a website.

Do not assume a test succeeded. To test SSH, confirm public addressing and routing, use the correct key and login name for the chosen AMI, and allow TCP/22 from your client IP. To test HTTP(S), install and start a web server first, allow its port, and verify the host firewall and application. Follow your organization's policies and account safeguards.

12 Understand the traffic flow

Suppose a user on the internet requests a web page served by `Learning-EC2`. Conceptually, a request travels from the client through the Internet Gateway toward the VPC, where routing and controls direct or permit it to the instance. Successful delivery requires all of the following:

- The EC2 network interface has a public IPv4 address (or another supported public addressing arrangement).
- The IGW is attached to `Learning-VPC`.
- The route table actually associated with the subnet has an applicable route to the IGW, here `0.0.0.0/0`.
- Network ACL rules allow the request and return traffic. NACLs are stateless, so relevant directions must be allowed.
- The instance's security group permits the intended inbound protocol/port and source. Security groups are stateful.

- The operating system firewall permits the traffic and the intended application is installed, running, and listening.

AWS maps the instance's public IPv4 address to its private IPv4 address for communication through an Internet Gateway. The instance itself uses its private address on its network interface. A public address without an IGW route is not enough; a route without a public address is also not enough for ordinary direct IPv4 internet communication. A subnet is considered public because its associated route table has a route to an IGW, not merely because an instance has a public address. Conversely, lacking public IPs alone does not define a subnet as private; its routing configuration does.

Conceptual internal example (not built in this lab): if a second instance were placed in another non-overlapping subnet in the same VPC, traffic between their private IP addresses could use the VPC local route. Security groups, NACLs, host controls, and application settings would still determine whether communication succeeds. This lab launches only one instance.

13 Troubleshooting

Check layers in order rather than assuming every failure is a security-group issue: addressing and placement; active route-table association and routes; IGW attachment; NACL and security-group rules; host firewall and service; then client-side configuration.

Table 4: Common symptoms and useful checks.

Problem	Possible cause	What to verify
Subnet CIDR rejected	Outside the VPC range or overlaps an existing subnet	VPC IPv4 CIDR and all existing subnet ranges; use a contained, non-overlapping range
IGW cannot be used	Not attached to the intended VPC	IGW attachment state and VPC association
Instance has no public IPv4	Auto-assignment disabled, launch configuration, or no public IP allocated	Instance networking details, subnet public-IP attribute, and launch network settings
Instance cannot reach internet	Missing default route, wrong association, or no public address	Active subnet route table, IGW attachment, route target, and instance addressing
External SSH fails	Routing, public addressing, SG, NACL, host firewall, SSH service, username, or key issue	Check each layer and confirm the client source IP is allowed
Website is unreachable	No server, wrong port, service not listening, missing rule, or routing issue	Application/service state, host firewall, SG/NACL, public address, and route
Route seems ineffective	Subnet uses another route table	Explicit subnet associations and the VPC's main route table
Internal works; external fails	Local routing exists but internet path/public address may not	Distinguish the local route from default route, IGW attachment, and public IPv4 configuration

14 Verification checklist

Use the console to verify actual configuration; do not treat this list as evidence that resources have already been created.

☐ VPC `Learning-VPC` exists and its IPv4 CIDR is `10.0.0.0/16`.

☐ Subnet `Learning-Public-Subnet` exists with CIDR `10.0.1.0/24`.

☐ The subnet is in the intended VPC and chosen Availability Zone.

☐ Route table `Learning-Public-RT` exists and has the VPC local route.

☐ Internet Gateway `Learning-IGW` is attached to the correct VPC.

☐ Route `0.0.0.0/0` targets the Internet Gateway.

☐ The custom route table is explicitly associated with the intended subnet.

☐ `Learning-EC2` is in the intended VPC and subnet.

☐ A public IPv4 address is present if direct IPv4 internet communication is intended.

☐ The security group contains only the inbound rules needed for the intended test.

☐ EC2 status checks have been reviewed.

☐ Connectivity is tested only after checking routing, addressing, security controls, host settings, and the service.

15 Exercises and knowledge checks

Answer the questions before reading the key.

1. Why is `10.0.1.0/24` a valid subnet inside `10.0.0.0/16`?
2. How many total IPv4 addresses are in a /24, and how many are available for AWS assignment in an ordinary AWS IPv4 subnet?
3. How many IPv4 addresses does AWS reserve in an ordinary IPv4 subnet? Why is the generic "subtract network and broadcast" host-count shortcut not the right AWS count?
4. What does the VPC local route accomplish, and what does it not guarantee?
5. Why does attaching an Internet Gateway not automatically make every subnet public?
6. What does `0.0.0.0/0` → `Internet Gateway` mean?
7. Why must `Learning-Public-RT` be associated with `Learning-Public-Subnet`?
8. Name the conditions required for an internet user to access a service on the EC2 instance.
9. Can two subnets in the same VPC use overlapping IPv4 CIDRs? Explain.
10. What is the difference between a route table and a security group?

15.1 Answer key

1. A /24 has a 24-bit network prefix and falls wholly within the larger /16 range. It is therefore a valid subset of the VPC's address space.
2. It has $2^{32-24} = 256$ total addresses. AWS reserves five in an ordinary IPv4 subnet, so 251 are available for assignment by AWS.
3. AWS reserves the first four and the last address. The familiar generic calculation subtracts two addresses, but AWS has additional reservations; for this ordinary /24, use 256 total and 251 AWS-assignable addresses.
4. The local route directs traffic for the VPC's local CIDR range within the VPC. It does not bypass security-group, NACL, or host filtering.
5. An IGW attaches at VPC level. A subnet needs an associated route table with a route to it; instances also need suitable public addressing and permitted traffic.
6. It is a default IPv4 route: destinations not covered by a more-specific matching route use the IGW target. The route does not itself grant access or assign a public IP.
7. A subnet's associated route table determines its paths. Without an explicit association it uses the main route table, which may not contain the intended IGW route.
8. The instance needs suitable public IPv4 addressing, the IGW attached to its VPC, a subnet-associated route table routing to the IGW, allowed NACL and SG traffic, permissive host settings, and an application listening on the requested port.
9. No. Subnet CIDR blocks in a VPC must not overlap, so each address maps unambiguously to one subnet.
10. A route table selects a network path/target for a destination. A security group filters permitted traffic for associated network interfaces/resources. Routes do not authorize connections; security groups do not create routes.

16 Quick Reference

Table 5: Resource purpose and the configuration to remember.

Resource	Job	Example / key configuration
VPC	Logically isolated network and IPv4 space	`Learning-VPC`: `10.0.0.0/16`
Subnet	Address range in one AZ	`Learning-Public-Subnet`: `10.0.1.0/24`
Route table	Directs packets by destination	`10.0.0.0/16` → `local`; `0.0.0.0/0` → `IGW`
Internet Gateway	VPC internet connection point	`Learning-IGW`, attached to the VPC
EC2 instance	Compute launched into a subnet	`Learning-EC2` in `Learning-VPC` / `Learning-Public-Subnet`
Security group	Allows traffic to/from associated network interfaces	Permit only required protocols, ports, and sources
Network ACL	Allows or denies traffic at subnet level	Check inbound and outbound rules; stateless

Relationship summary:

- A subnet is associated with one route table; a route table can serve multiple subnets.
- An Internet Gateway is attached to a VPC; a route table can contain a route targeting it.
- An EC2 instance is launched into a subnet.
- A security group controls permitted traffic to and from an associated network interface/resource.
- A route table determines paths, not access permissions.

Example configuration:

- VPC CIDR: `10.0.0.0/16`; subnet CIDR: `10.0.1.0/24`.
- Local route: `10.0.0.0/16` → `local`; internet route: `0.0.0.0/0` → `Internet Gateway`.
- EC2 placement: `Learning-VPC` and `Learning-Public-Subnet`.

17 References

The following official AWS documentation was checked for the technical behavior and console workflows described here. AWS may update documentation and console labels; consult the current version while working.

[1] How Amazon VPC works — VPC concepts and logical network overview.

[2] Subnet CIDR blocks — CIDR sizing, overlap restrictions, and five reserved IPv4 addresses.

[3] Create a VPC — VPC creation workflow.

[4] Create a subnet — subnet creation and Availability Zone selection.

[5] Subnet route tables — routes, main route table, and subnet associations.

[6] Enable internet access using an internet gateway — attach the gateway, configure routes, and public IPv4 requirements.

[7] Manage IPv4 addresses for EC2 instances — public/private IPv4 assignment.

[8] Security groups — resource/network-interface traffic filtering.

[9] Network ACLs — subnet-level traffic filtering.

[10] EC2 launch configuration parameters — instance, subnet, public IP, and security-group network settings.